ALGEBRAIZATION OF A FORMAL CLOSED SUBSPACE OF AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10168-10191, immediately following the statement at 10145-10166.

Editorial note. Same-name pair note: this is the more general (algebraic spaces) side; the schemes side is not counted beside it.

Editorial note. The Grothendieck existence theorem and its reformulation are same-chapter navigation pointers recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Grothendieck's existence theorem for algebraic spaces. chapter theorem theorem-grothendieck-existence, cited in the proof. Produces the algebraization of the compatible system of ideal sheaves on the thickenings.
- Reformulation of existence as an equivalence of categories. chapter remark remark-reformulate-existence-theorem, cited in the proof. Gives the form of the theorem applied to the ideals defining the formal closed subspace.
- Closed subschemes and properness after algebraization. used at the end of the proof (Z proper over S). Reads off that the algebraized object is a closed immersion and proper over S.

Source reference. Original lemma 08B5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9644–9644. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08BF: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10085–10085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08B2: focused-seeds/geometry/S9-spaces-algebraize-formal-closed-subscheme-spaces/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to an ideal I. Write $S = \mathrm{Spec}(A)$ and $S_n = \mathrm{Spec}(A/I^n)$. Let $X \to S$ be a morphism of algebraic spaces that is separated and of finite type. For $n \geq 1$ we set $X_n = X \times_S S_n$. Suppose given*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

a commutative diagram

$$\begin{array}{ccccccc} Z_1 & \longrightarrow & Z_2 & \longrightarrow & Z_3 & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \downarrow & & \\ X_1 & \xrightarrow{i_1} & X_2 & \xrightarrow{i_2} & X_3 & \longrightarrow & \ldots \end{array}$$

of algebraic spaces with cartesian squares. Assume that

(1) *$Z_1 \to X_1$ is a closed immersion, and*
(2) *$Z_1 \to S_1$ is proper.*

Then there exists a closed immersion of algebraic spaces $Z \to X$ such that $Z_n = Z \times_S S_n$ for all $n \geq 1$. Moreover, Z is proper over S.

Proof. Let's write $j_n : Z_n \to X_n$ for the vertical morphisms. As the squares in the statement are cartesian we see that the base change of j_n to X_1 is j_1. Thus Limits of Spaces, Lemma 08B2 shows that j_n is a closed immersion. Set $\mathcal{F}_n = j_{n,*}\mathcal{O}_{Z_n}$, so that $j_n^\sharp$ is a surjection $\mathcal{O}_{X_n} \to \mathcal{F}_n$. Again using that the squares are cartesian we see that the pullback of $\mathcal{F}_{n+1}$ to X_n is $\mathcal{F}_n$. Hence Grothendieck's existence theorem, as reformulated in Remark 08BF, tells us there exists a map $\mathcal{O}_X \to \mathcal{F}$ of coherent $\mathcal{O}_X$-modules whose restriction to X_n recovers $\mathcal{O}_{X_n} \to \mathcal{F}_n$. Moreover, the support of $\mathcal{F}$ is proper over S. As the completion functor is exact (Lemma 08B5) we see that $\mathcal{O}_X \to \mathcal{F}$ is surjective. Thus $\mathcal{F} = \mathcal{O}_X/\mathcal{J}$ for some quasi-coherent sheaf of ideals $\mathcal{J}$. Setting $Z = V(\mathcal{J})$ finishes the proof. □